

Geographical Information System Based Assessment of Small Hydropower Potential in South-Eastern Nigeria: A Case Study of Abia State

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Abstract. Small hydropower technology has gained traction in the Nigerian energy and power ecosystem owing to incentives and reforms aimed at increasing Nigeria's energy mix for sustainable development. Utilizing these opportunities through harnessing SHP potentials has not made it to the front burner during policy formulations and implementations in South-Eastern Nigeria despite the availability of water bodies and waterlines in the region. This paper focuses on the potentials of small hydropower in Abia state and utilized ArcGIS software to conduct spatial analysis using map data overlayed by shapefiles of water bodies, waterlines, road networks and Land use, Land Cover data (LULC). Multiple ring buffers were created for various proximities around the waterbodies and waterlines and Normalized Difference Vegetation Index calculations were done to determine suitability areas for small hydropower schemes after reclassification of the data. The analysis revealed suitability areas in Osisioma Ngwa and Obingwa Local Government areas with suitable elevations and hydraulics data for run off the river schemes and siting of hydropower plants within a multiple ring buffer distance between 200m to 5km from the waterlines and roads, having a weighted score between 33-66 with NDVI range of -0.018 – 0.015 indicating the presence of water bodies and built-up areas around the water bodies with NDVI range of 0.015 – 0.14 and a weighted score within the range of 11-16 This revelation also encourages hybridization of renewable energy technology using pumped hydro storage to improve the reliability and affordability of mini-grid solutions in Abia state and Nigeria at large.

Introduction

One of Nigeria's six geopolitical zones, the southeast is renowned for its rich cultural legacy, spirit of entrepreneurship, and economic activity. Five states make up the region: Abia, Anambra, Ebonyi, Enugu, and Imo. Trade and commerce, manufacturing, agriculture, banking services, education and healthcare services, entrepreneurship, entertainment, transportation, and innovation and technology are among the region's significant economic activities [1]

Due to the predominance of the Igbo ethnic group, which is renowned for its strong commerce and business abilities, the southeast is sometimes referred to as the "Igbo land." Onitsha and Aba are two well-known commercial hubs where a variety of items are purchased and sold [2]. Markets in these places are thriving and draw vendors from all across Nigeria and its neighbors.

Manufacturing and industry are concentrated in the southeast, especially in places like Aba. The region's small and medium-sized businesses manufacture a variety of items, including textiles, footwear, leather goods, plastics, and electronics [3]. High-quality goods are always associated with the "Aba Made" trademark. In the southeast, agriculture plays a significant role in the economy. The production of crops including yam, cassava, oil palm, and numerous fruits and vegetables is well-known in the area. Agribusiness and commercial farming are also gaining in popularity [4].

With several banks, microfinance organizations, and other financial services businesses operating in the region, the financial industry is expanding in the southeast. Cities like Owerri and Enugu are

developing into major financial and commercial hubs [5]. Universities, polytechnics, and hospitals are only a few of the educational and medical facilities located in the southeast. These industries generate income and provide up job possibilities. The Igbo people's entrepreneurial spirit has resulted in the formation of numerous small and medium-sized firms (SMEs) in a variety of industries, considerably boosting regional economic activity [6].

Due to urbanization and rising population, the construction and real estate industries have experienced significant expansion in the south-east [7]. The construction of commercial and residential structures is a part of this increase [8]. The area is a thriving center for Nigeria's entertainment sector, especially Nollywood, the country's film industry. Numerous eminent actors, directors, and filmmakers have come from the southeast. Due to its central location in Nigeria, extensive road systems, and closeness to other nations, the southeast serves as a vital center for logistics and transportation [9]. Numerous logistics enterprises, markets for the distribution of commodities, and transport companies are located in the region. In the southeast, startups and tech centers are flourishing, and there is a rising interest in innovation and technology. This demonstrates the region's dedication to broadening its economic base outside established industries [10].

In spite of these commercial activity, it's important to note that the area still faces difficulties including a lackluster power supply, infrastructural gaps, security issues, and political problems [11]. The region's economic growth depends on overcoming these obstacles and enhancing the electricity supply that powers these economic activities.

This paper seeks to provide the much-needed background and understanding of alternative and renewable sources of generating electricity in Abia state to meet the enormous existing demand for electricity without reliance on the national grid with focus on the utilization of small hydropower potentials in the state.

Overview of Abia State Power Sector

Abia State, like many other states in Nigeria, faced challenges. Nigeria, in general, has struggled with an unreliable and insufficient power supply due to a variety of issues, including inadequate infrastructure, funding, and mismanagement in the power sector [12].

The power supply situation in Abia State, as in other parts of Nigeria, was characterized by frequent power outages and load shedding [13]. Many areas in the state relied on backup generators and inverters to cope with the irregular power supply [14]. Below are overview of the status of the various value chains of the power industry in Abia state;

Generation: Abia State lacks a power plant capable of supplying the region's energy demands. The national power grid is the only source of electricity for the state [15]. Throughout Nigeria, electricity is largely produced by power plants, which subsequently distribute the energy to other states, including Abia [16,17]. Some power generation projects in Abia state include;

Aba integrated power project (Aba IPP): The building and management of a power plant in Aba, Abia State, are handled by Geometric Power Limited as part of the Aba Integrated Power Project. The Aba Ring-Fenced Industrial and Export Processing Zone, a designated region in Aba, Abia State, Nigeria, may be supplied by the power plant, which is built to generate electricity to fulfill the energy demands of the commercial and industrial sectors in Aba and its surroundings [18]. The goal of this industrial and export processing zone is to boost exports, support industrialization, and advance economic growth in a particular geographic area.

Alaoji power plant: The Alaoji power station is a working power plant with various units, some of which are not currently in operation. It has a minimum capacity of 1068 megawatts (MW). The initial 540 MW phase was finished and put into service in March 2015 [15].

Distribution: The Enugu Electricity Distribution Company (EEDC), which manages the distribution of electricity in the Southeast of Nigeria, is responsible for distributing Electricity to the Abia state's domestic, commercial, and industrial consumers [19].

Due to the national grid's unpredictable power supply, Abia State households and businesses have increasingly explored alternative energy sources including solar energy and generators to satisfy their energy demands [20]. As it is with the rest of the country, Abia State's electricity industry has to deal with issues including poor infrastructure, power theft, significant transmission and distribution losses, and budgetary limitations. These problems have impacted on the supply's consistency and quality.

Renewable Energy Potentials in Abia State

The need to diversify the energy mix, lower greenhouse gas emissions, and supply sustainable and dependable energy sources is what motivates the usage of renewable energy in Abia State and other regions of Nigeria. [21]. In Abia State, several projects and sources of renewable energy are being investigated;

Solar Energy: According to the Global Solar Atlas, Abia State receives a lot of sunshine throughout the year, with a Global Horizontal Irradiation (GHI) of 1684.6 kWh/m²/yr and a specific PV power output (PVOUT) of 1342.5 kWh/kWp [22], making solar energy a practical source of renewable energy. Solar power may be utilized to power buildings such as residences, companies, and public institutions both on and off the grid.

Biomass Energy: Biomass Energy: Another form of renewable energy in Abia State is biomass energy, notably from organic and agricultural waste. Cooking, heating, and the production of power are all possible with biomass [23]. According to data from United Nations Food and Agricultural Organization UN-FAO, Abia state has an energy crop production capacity of 28.604Gtonnes/yr [24]

Waste-to-Energy: There are initiatives to use anaerobic digestion and the creation of biogas to turn organic waste into electricity [25]. This can serve as a source of renewable energy and aid in trash management. Municipal solid waste (MSW) in Nigeria was evaluated at the state level by Tosin et al. in 2017. The research demonstrated that, depending on the state's capability for producing trash, the power generation potential for the various states in Nigeria ranged from 31 to 205 MW. The potential for the nation to generate energy from MSW annually was projected to be 26,744 GWh/year, with 89% of the states having adequate production capacity at the regulation minimum of 50 MW [26]. From another study conducted in within Umuahia Metropolis, Abia state has great potentials to generate electricity from Household waste, as this records an average of 0,53kg/day/household [27].

Hydropower: The geography and river network of Abia State offer the possibility for minor hydropower installations. These projects utilize the movement of water in rivers and streams to produce power [28]. According to a recent study, the absence of inland water and inundated land sites places a limitation to the capacity of hydropower scheme possible within the state. The study however recommended run-off-river schemes to power micro and Pico turbines for power generation in locations such as these [29,30].

Wind Energy: Due to the fluctuating wind patterns, solar and hydroelectric are often more prevalent than wind energy in Abia State. Studies suggest despite the low potentials of wind energy in Abia state owing to low elevation and wind speed, there are still opportunities for small scale investments in about 108 potential sites in Abia south covering a distance of 270Km² [31]. However, to find suitable sites for wind energy projects, wind resource assessments evaluations could be carried out.

Small Hydropower Potentials in Abia State

The terrain of Abia State in Nigeria and the existence of several rivers and streams both contribute to the state's small hydropower potential [32]. The movement of water in rivers and streams may be used to power small hydropower plants, and Abia State possesses several features that make such projects feasible. Studies suggest the following locations have potential features for possible siting of small hydro projects; Arochukwu, Umuahia, Obingwa, Osisioma-Ngwa, Ohafia, Bende and Isiukwuato. These assumptions are based on the terrain, and network of streams and rivers abundant in these areas. The figure below shows the map of Abia state showing waterlines and rivers

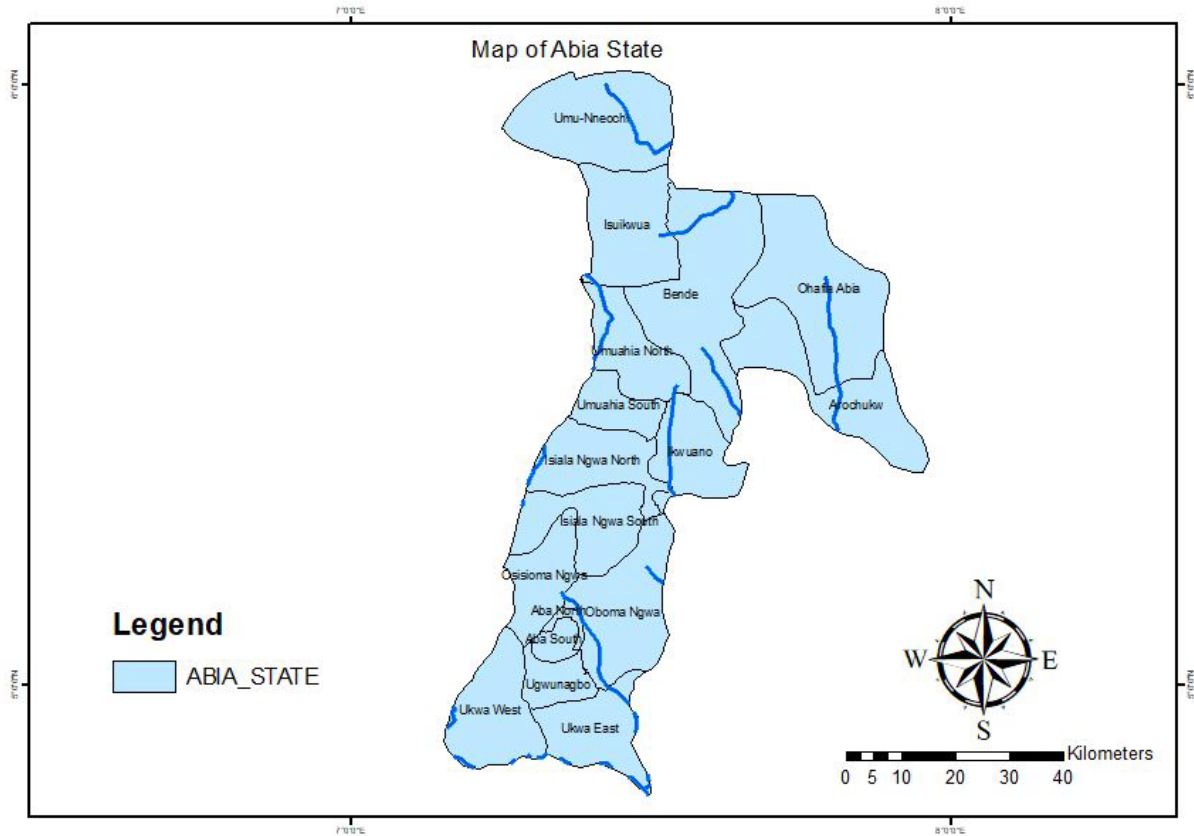


Fig.1. Map of Abia state showing waterlines. *Source: (ESRI, 2020).*

Materials and Methods

This study assesses these potentials using a geospatial tool to determine the suitability areas within these locations mentioned above for small hydro generating plants using run-of-river schemes and the potentials for these schemes to support the hybridization of other renewable energy conversion technologies with small hydro or pumped hydro schemes for a medium to large-scale power production to meet energy demands of the various consumers within Abia state for sustainable development as contained in the 2019 report of United Nations Development Organization (UNIDO) [33].

The ArcGIS, a software developed by the Environmental Systems Research Institute (ESRI) offers a wide range of tools and capabilities enabling users to organize, examine, and display geographic data for a variety of purposes [30]. The study uses two spatial analytical methods namely; Normalized Difference Vegetation Index (NDVI) calculations and Weighted Overlay analysis to analyze map files containing Land Use Land Cover data, Water bodies and water areas raster data overlaid on a map shape file (ABIA.shp) extracted using a clip tool from NGA_adm0, 1, 2 map files obtained from ESRI map datasets.

The Normalized Difference Vegetation Index (NDVI): NDVI is used to quantify vegetations by calculating the vegetation that is strongly reflected (near-infrared, NIR) and the vegetation that is absorbed (RED) [34], its value ranges from -1 to +1, with negative values indicating high likelihood of water and positive values indicating high likelihood of dense green leaves, which are indicative of high temperature and tropic rain-forest areas. Values from; -0.018 to 0.015 indicates area characterized by water, and 0.015 to 0.14 indicates built-up area. NDVI calculation is represented with the formula given below;

$$NDVI = \frac{NIR - RED}{NIR + RED} \quad (1)$$

Weighted overlay calculation: The weighted overlay analysis tool is used to display low – high potential areas on a scale of 1-9 [35]. LULC raster files are reclassified and overlaid with road_r and water_r raster files in given percentages totalling 100% to show suitability areas for small hydropower plants in Abia state. The weighted score is calculated as shown below;

$$W_{score} = \sum_m^n C_i * W_i . \quad (2)$$

Where;

i = (roadlines, waterlines, multiple ring buffer with distances from road etc);

W_{score} is overall weighted overlay score;

C_i is criteria score of i;

W_i is weight value of criteria i.

Suitability analysis: The Map algebra, a spatial analysis tool is used to perform raster calculations to determine suitability area SA and is represented by the equation below,

$$S_A = LULC * W_i. \quad (3)$$

Where LULC is the reclassified Land Use Land Cover raster data.

Flow chart: The following steps are taken to carry out spatial analysis on the ArcGIS platform to determine locations suitable for siting small hydropower schemes in Abia state.

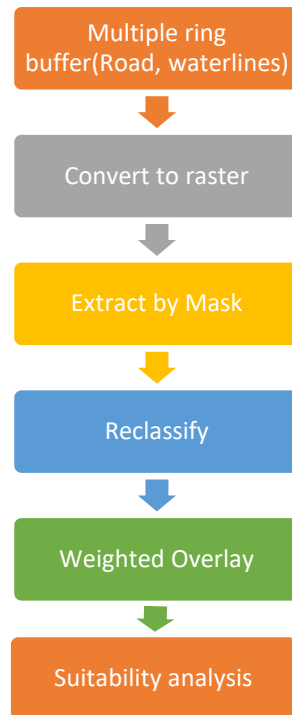


Fig. 2. ArcGIS spatial analysis flow chart

Assessing Small Hydropower Potentials in Abia State

Map datasets for NGA_roads and NGA_waterlines were downloaded from ESRI portal [36]. The map of Abia state was extracted from the NGA_0, 1, 2 shape files with the road and waterlines extracted using clip, a spatial analyst tool on ArcGIS.

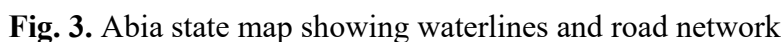


Table 1. Reclassification of road and waterline raster data with output colors.

Classification	Distance[m]	Colors (Road_r)	Colors(waterlines_r)
9	200	Dark Umber	Ultra blue
7	200-700	Tuscan red	Yogo blue
5	700-1200	Pointsettia red	Cretan Blue
3	1200-3000	Medium coral light red	Lapiz Lazuli Blue
2	3000-5000	Rose Quartz	Sungilite sky blue

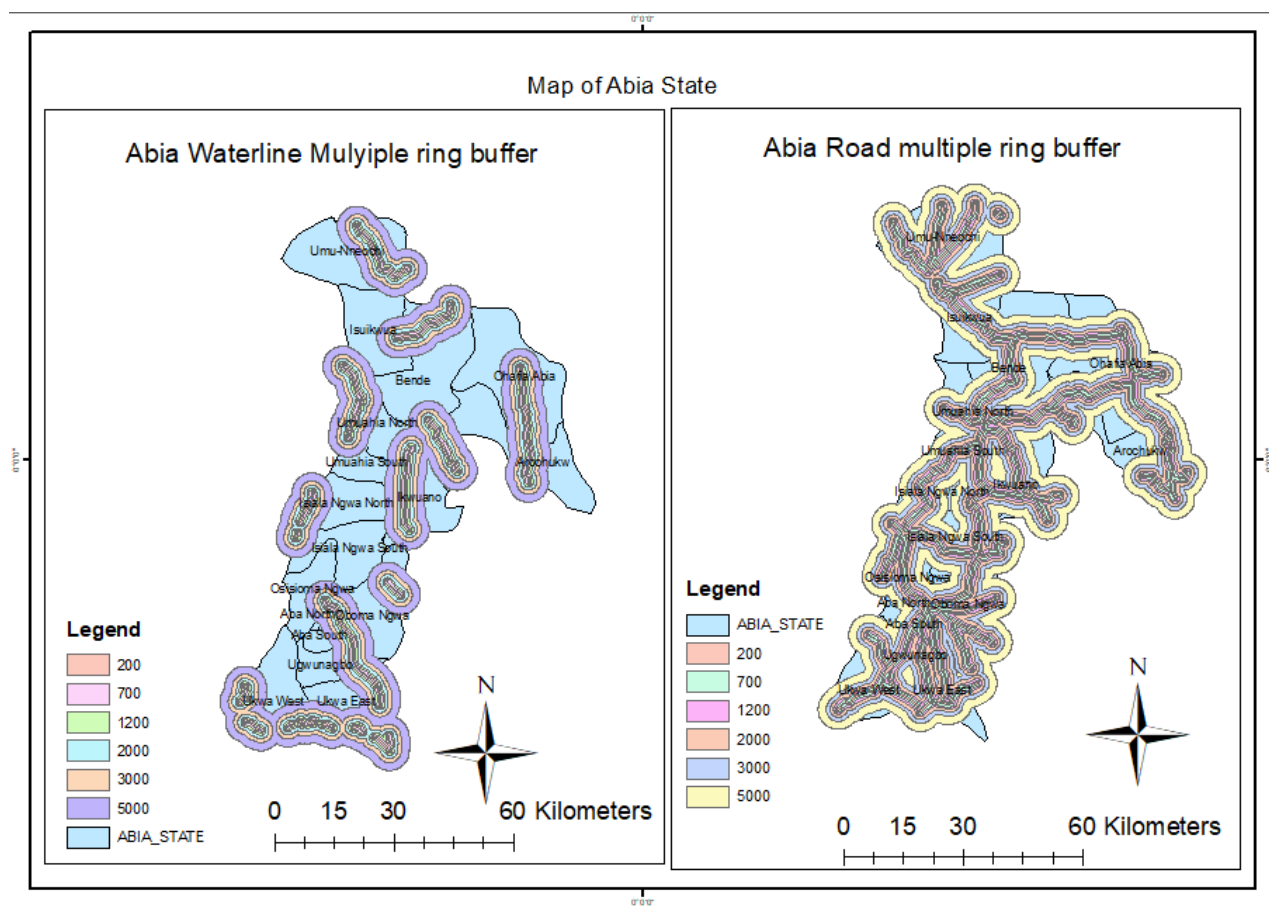


Fig. 4. GIS Map showing Multiple ring buffers for Abia state waterlines and roads

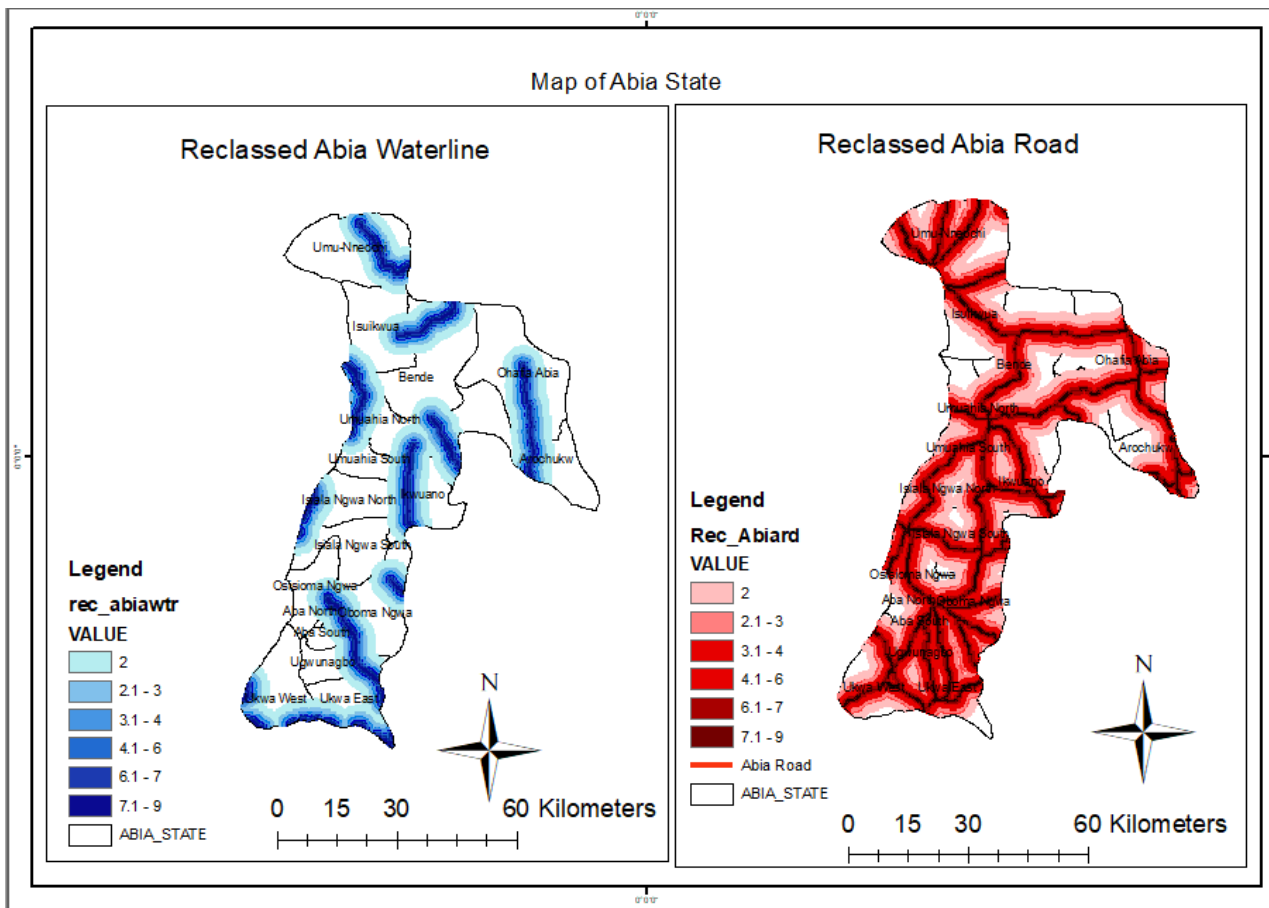


Fig. 5. GIS Map of Abia state showing reclassified waterlines and road raster

Table 2. Reclassification of Land use Land Cover (LULC) raster with output colors shown

RECLASSIFIED LULC

Legend

Reclass_LULC Value	Description
1	Dark Green
1.1 - 2	Green
2.1 - 3	Light Green
3.1 - 5	Yellow
5.1 - 7	Pink
7.1 - 9	Blue
ABIA_STATE	Light Blue

Scale: 0 5 10 20 30 40 Kilometers

Weighted overlay calculation: The reclassified maps for LULC, road_r and Waterline_r were overlaid using various weighted scores of 40%, 30% and 30% respectively totaling 100%

Results and Discussions

The figures below show the results of the weighted overlay and map algebra calculations done using the ArcGIS platform to determine suitable areas for siting small hydropower projects in Abia state.

Fig. 8 below shows results from the suitability analysis revealing potential sites across four local government areas, namely; Osisioma-Ngwa, Aba North, Aba South and Obingwa Local government areas as could be clearly seen by the lapis lazuli blue coloration for water bodies with a class weight between 36-63. Other potential areas include; Ohafia and Umuahia North L.G.As. The possibility of siting these schemes in areas accessible by road and within built-up areas can be seen from the orange to lemongrass and red colorations. Other locations like Umunneochi, Umuahia south, Ikwuano and Isiala Ngwa North also shows possible habitations along the waterlines, however there are just little waterbodies in these areas, making such investment on power generation not viable.

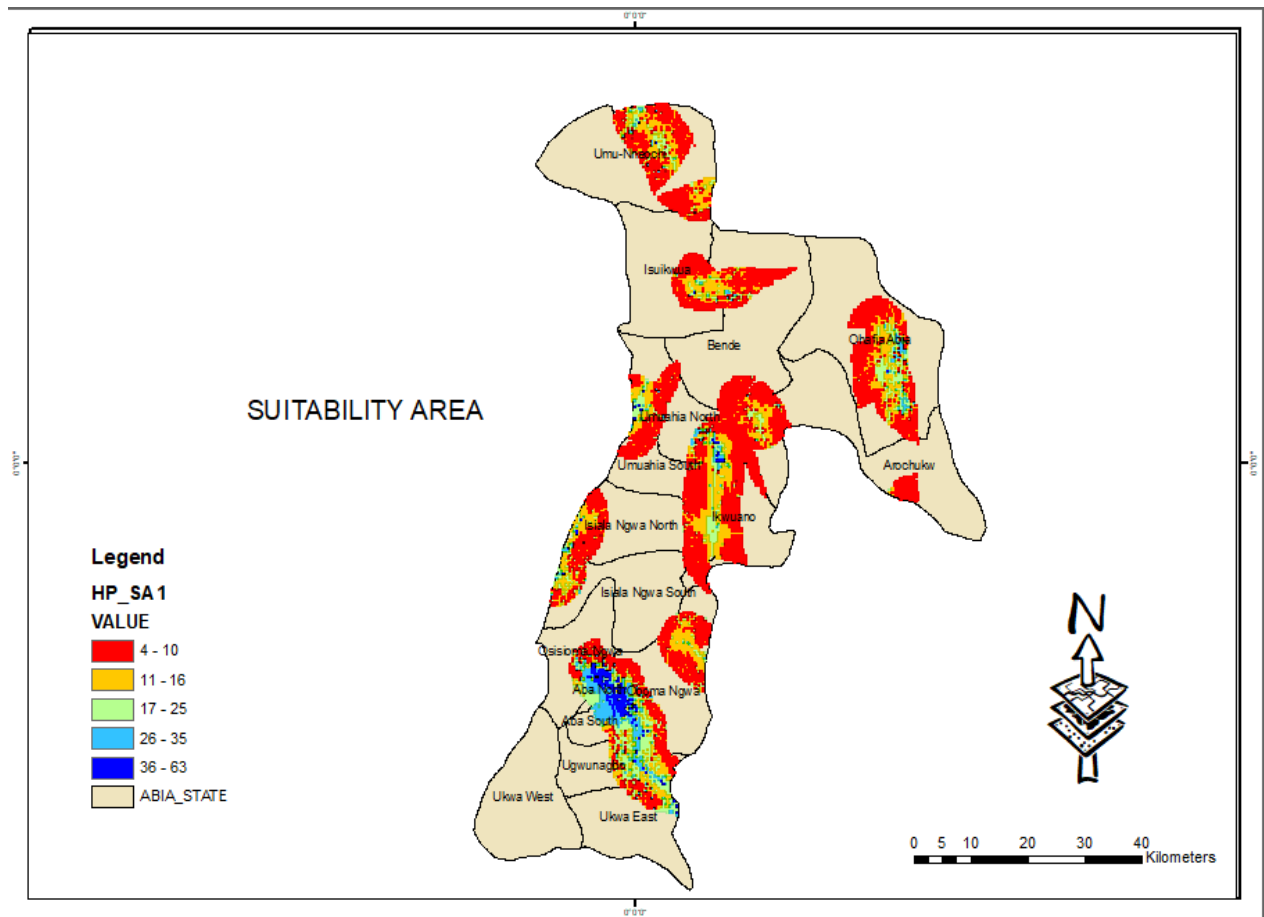


Fig. 8. Map of Abia state showing suitability areas for small hydropower generation

Conclusion and Recommendation

This study reveals the possibility to harness available potentials for small hydropower within Abia south and Abia central senatorial districts for power generation and irrigation. This is shown by the availability of suitability areas in Osisioma Ngwa and Obingwa Local Government Areas having a weighted score between 33-66 with NDVI range of -0.018 –0.015 indicating the presence of water bodies and built-up areas around the water bodies with NDVI range of 0.015 – 0.14 and a weighted score of 11-16. These untapped potentials could be harnessed to meet the energy demands of the industrial clusters and small and medium scale business with Aba metropolis and serve as a means to provide water supply for large scale farming. The study hereby also recommends the construction of pumped hydro storage facilities in Osisioma-Ngwa and Obingwa Local government areas, to serve as a backup storage for planned mini-grid projects by the Rural Electrification Agency through the Nigerian Electrification project and encourages project developers to explore this option as a means to reduce overall capital expenditure. Levelized cost of energy and make electricity from mini-grids affordable to low-income communities in Abia state. Furthermore, efforts should be made by the government to conduct feasibility studies in these areas to determine the possible capacity of small hydropower plants and the scheme that will be suitable for these locations based on available hydrological and hydraulic data, such as head (m) and Volumetric flow rate of the water $Q(m^3/s)$ as these determine the type of hydro turbines to install and the civil works required.

References

- [1] Profile of South-Eastern Nigeria and Description of the Study Area Profile of south-eastern Nigeria. (2001).
- [2] Ohadike, D. C. Igbo Culture and History.pdf. Preprint at (1960).
- [3] Chinwuba, P. Determinants of Participation in Nonfarm Economic Activities in South East Nigeria : A Tobit Analysis Approach. 5, 102–109 (2015).
- [4] World Bank. Nigeria Socio Economic Assessment 1. World Bank 3–17 (2011).
- [5] Umezina, C. Health and economic growth in South East, Nigeria. African Research Review 10, 111 (2016).
- [6] Odoh, N. E. & Nwibo, S. U. Socio-Economic Determinants of Rural Non-Farm Households Income Diversification in Southeast Nigeria. International Research Journal of Finance and Economics-Issue 164, 116–128 (2017).
- [7] Amaechi, C. M. & Muoh, O. U. Women and development in Africa: Urbanization and the role of Umuada in Igboland, Southeast Nigeria. Asian Women 34, 85–107 (2018).
- [8] Obi-Ani, N. A. & Isiani, M. C. Urbanization in Nigeria: The Onitsha experience. Cities 104, 102744 (2020).
- [9] Jelilov, G. & Iheoma, H. The Impact Of Industrialization On Economic Growth: The Nigeria Experience (2000-2013) Entrepreneurial Trade Fair (Turkmenistan) 2012 View project Entrepreneurial Trade Fair (Turkmenistan) 2010 View project THE NIGERIA EXPERIENCE (2000-2013). British Journal of Advance Academic Research 5, 11–20 (2016).
- [10] Unizik journal of business. 1, 81–92 (2018).
- [11] Idika, U. Factors that Hinder Effective Domestic Energy Management Practices within Households of Urban and Rural Areas of Abia State. 5, 1–16 (2018).
- [12] Diji, D. A Critical Review of the Nigerian Energy Scenario. IOSR Journal of Electrical and Electronics Engineering 5, 55–67 (2013).
- [13] Onwumere, J. C., Amaghionyeodiwe, C. A. & Ndukwe, E. C. Effect of epileptic power supply on the investment and performance of bakeries in Abia State, Nigeria. Nigeria Agricultural Journal 50, 92–99 (2019).
- [14] REA. Nigeria Electrification Project (NEP). (2023).
- [15] Okedu, K. E., Sarah, K., Kayoade, I. & Roland, U. Impact of the Independent Power Producers and National Integrated Power. 20, 147–170 (2018).
- [16] Facts, B. Engineering The Profile of Energy Generation and Consumption in Nigeria Department of Pure and Industrial Chemistry , Nnamdi Azikiwe Department of Chemical Engineering , Nnamdi Azikiwe University ,. 194–201 (2016).
- [17] Elizondo, E. et al. Productive Use of Energy – PRODUSE Measuring Impacts of Electrification on Small and Micro-Enterprises.
- [18] Ukoima, K. & Agwu, E. O. Review of the Impact of Electricity Supply on Economic Growth : A Nigerian Case Study Review of the Impact of Electricity Supply on Economic Growth : A Nigerian Case Study. IOSR Journal of Electrical and Electronics Engineering 14, 28–34 (2019).
- [19] Amuji, H. O., Nwachi, C. C., Nicholas, N., Mbachu, J. C. & Owolabi, W. T. Correlation Analysis of Enugu Electricity Distribution Company ' s Electricity Bill. 10–19 (2023) doi:10.5281/zenodo.7765101.
- [20] Clinton, D. & Chinago, A. B. State of power supply in Nigeria, the way out. International Journal of Development and Sustainability 7, 435–447 (2018).

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- [21] Igbinovia, F. An overview of Renewable energy potential in Nigeria: Prospects, Challenge and the Way forward. *Energetika Journal* 46, 570–579 (2014).
- [22] Njoku, M. C. et al. GLOBAL SOLAR RADIATION UTILIZATION IN SOUTH-EAST OF NIGERIA. 1141–1152 (2022).
- [23] Ramanjaneyulu, G. & Reddy, B. R. Emerging trends of microorganism in the production of alternative energy. *Recent Developments in Applied Microbiology and Biochemistry* 275–305 (2018) doi:10.1016/B978-0-12-816328-3.00021-0.
- [24] Ukoba, M. O. et al. Geographic information systems (GIS) approach for assessing the biomass energy potential and identification of appropriate biomass conversion technologies in Nigeria. *Biomass Bioenergy* 170, 106726 (2023).
- [25] Joardder, M. U. H., Halder, P. K., Rahim, M. A. & Masud, M. H. Solar pyrolysis: Converting waste into asset using solar energy. *Clean Energy for Sustainable Development: Comparisons and Contrasts of New Approaches* 213–235 (2017) doi:10.1016/B978-0-12-805423-9.00008-9.
- [26] Somorin, T. O., Adesola, S. & Kolawole, A. State-level assessment of the waste-to-energy potential (via incineration) of municipal solid wastes in Nigeria. *J Clean Prod* 164, 804–815 (2017).
- [27] Chima, I. K. & Nwabinye, V. Economic Analysis of Household Waste Generation, Disposal and Management in Umuahia Metropolis, Abia State, Nigeria. *International Journal of Health Economics and Policy* 2, 47 (2017).
- [28] Ohunakin, O. S., Ojolo, S. J. & Ajayi, O. O. Small hydropower (SHP) development in Nigeria: An assessment. *Renewable and Sustainable Energy Reviews* 15, 2006–2013 (2011).
- [29] Odiji, C. et al. Small hydropower dam site suitability modelling in upper Benue river watershed, Nigeria. *Appl Water Sci* 11, (2021).
- [30] Oyinna, B. et al. Assessing small hydropower sites in Nigeria for sustainable development using ArcGIS. *Energy Reports* 10, 2889–2898 (2023).
- [31] CHIEMELU, N.E., NKWUNONWO, U., OKEKE, F.I. & OJINNAKA, O.C. Geospatial Evaluation of Wind Energy Potential in the South-East and South-South Sections of Nigeria. *International Journal of Environment and Geoinformatics* 6, 244–253 (2019).
- [32] Nkemdirim, V.U., Alozie, G. C. & Felix, I. Spatial Analysis of the Quality of Groundwater Supply: Evidence from Abia State. *American Journal of Water Resources* 4, 83–90 (2016).
- [33] UNIDO. Incentive Policies for Small Hydropower Development. 1–8 (2019).
- [34] Ukoba, M. O. et al. Optimal sites for agricultural and forest residues energy conversion plant using geographic information system. *Heliyon* 9, e19660 (2023).
- [35] Asadi, M., Pourhossein, K. & Mohammadi-Ivatloo, B. GIS-assisted modeling of wind farm site selection based on support vector regression. *J Clean Prod* 390, (2023).
- [36] ESRI Geographical Information System, 2020.